

Doruk Üstündag

dorukustundag5@hotmail.com | +90 506 855 7005

EDUCATION

IZMIR INSTITUTE OF TECHNOLOGY

BACHELOR OF SCIENCE IN MATHEMATICS

Expected Jun 2027 | Izmir, Turkey

High Honors (2nd Semester)

Honors (3rd Semester)

Honors (4th Semester)

Cum. GPA: 3.41 / 4.00

RESEARCH

IZTECH MATHEMATICS DEPARTMENT | UNDERGRADUATE

RESEARCHER

2024 – Present | Izmir Institute of Technology

Working with Assoc. Prof. Haydar Göral

PROJECTS

GRADUATION PROJECT | IZMIR INSTITUTE OF TECHNOLOGY

2025 – Present

Working on "Zeta Functions in Algebraic Structures" under the supervision of Assoc. Prof. Haydar Göral.

UNDERGRADUATE RESEARCH | IZMIR INSTITUTE OF TECHNOLOGY

2024 – 2025

Worked on "Combinatorial Interpretation of Power Sum Polynomials and Various Divisibility Properties" with Deniz Göksu (IZTECH) under the supervision of Assoc. Prof. Haydar Göral.

HIGH SCHOOL RESEARCH | TED IZMIR COLLEGE

2020 – 2022

Worked on "Investigation of Breast Cancer Diagnosis Using the Stress-Strength Reliability Model with a Fuzzy Approach" for TUBITAK 2204-A High School Science Projects Contest.

PUBLICATIONS

[1] D. Göksu and D. Üstündag. Combinatorial interpretation of power sum polynomials and various divisibility properties. *Preprint*.

RESEARCH INTERESTS

Computational Number Theory

Sums of Powers and Faulhaber's Polynomials

Analytic Number Theory

Algebraic Number Theory

Harmonic Numbers

SOCIETIES

2025 – Present

IZTECH Arithmetic | Co-President

2024 - 2025

IZTECH Mathematics Society | President

2023-2024

IZTECH Environment Society | Academic

Organisation Coordinator

IZTECH Radio Society | Radio Host

2017 - Present

Turkish Education Association | Fellow

AWARDS

- 2025 - Best Young Researcher Talk Certification - Izmir Mathematics Days (Ege University)
- 2023 - "Scientist Training Programme" Diploma - Turkish Education Association
- 2021 - Winners of the People's Vote - Earthbeat Film Challenge
- 2020 - Distinction Medal - Waterloo University

COMMUNITY SERVICE

Turkish Education Association-IZTECH Project Supervisor

Turkish Education Association Volunteer

SKILLS

PROGRAMMING

- 4+ years: Python
- 2+ years: SageMath
- 1+ years: Matlab
- 0+ years: R

GIVEN TALKS

- Algebraic Structures: Magmas to Groups @ Algebra 0 Seminar
- Introduction to Group Theory and Presentations of Finite Groups @ MATH200
- p -adic Metrics and the Infinitude of Primes @ MATH200
- A Computational Approach on Faulhaber's Polynomials and Their Divisibility @ IZTECH Number Theory Workshop
- Pozitif Tam Sayı Kuvvet Toplamlarının Kapalı Formları ve Kombinatorik Özellikleri @ National Mathematics Symposium
- Pozitif Tam Sayı Kuvvet Toplamlarının Kapalı Formları ve Kombinatorik Özellikleri @ Izmir Mathematics Days
- The Mathematics of Power Sums: History, Structure, and Applications @ Yozgat Bozok University (04 November)
- Harmonic Numbers and their p -Adic Structure @ IZTECH Arithmetic Seminar Series (11 November)

ORGANIZED EVENTS

- IZTECH Arithmetic Seminar Series
- BILFEST'25
- IZTECH Number Theory Workshop
- PiDay'25
- Mathematics and Chess Workshop
- IZTECH Algebra Reading Group: Algebra 0
- Mathematics Department Book Donation Event
- IZTECHBEE'24 (Integral Bee)
- BILFEST'24

ADDITIONAL INFORMATION

- Languages: Turkish (Native), English (Fluent)
- Available for research collaborations and academic projects

LINKS

LinkedIn: **dorukustundag**

ResearchGate: **Doruk Ustundag**

IZTECH Arithmetic: **iytearitmetik**